Jack Chen-An Chou

CONTACT INFORMATION

School of Mathematics
University of Minnesota
Minneapolis, MN 55455

E-mail: chou0188@umn.edu

Website: https://jackchoumath.github.io/

EDUCATION

University of Minnesota, Twin Cities, Minneapolis, MN — 2026 - Present
Ph.D. in Mathematics

University of Florida, Gainesville, FL — 2024 - 2026
M.S. in Mathematics, Advisor: Zachary Hamaker

University of California, San Diego, La Jolla, CA — 2020 - 2024
B.S. in Mathematics, Advisor: Brendon Rhoades

RESEARCH INTERESTS

Algebraic combinatorics: Schubert calculus

PREPRINTS

[1] *Castelnuovo-Mumford regularity of skew-symmetric matrix Schubert varieties*
arxiv.org/abs/2609.24957

[2] *Coxeter and Schubert combinatorics of μ–involutions*
with Zachary Hamaker
arxiv.org/abs/2604.03379

[3] *A positive combinatorial formula for the double Edelman–Greene coefficients*
with Tianyi Yu
arxiv.org/abs/2512.12185

PUBLICATIONS

[4] *Newton polytopes of fireworks Grothendieck polynomials*
with Linus Setiabrata
arxiv.org/abs/2508.09107
Accepted in *Combinatorial Theory*

[5] *Asymptotically maximal Schubitopes*
with Linus Setiabrata
arxiv.org/abs/2512.04053
Bulletin of the London Mathematical Society, Volume 58, Issue 7

[6] *Grothendieck polynomials of inverse fireworks permutations*
with Tianyi Yu
arxiv.org/abs/2403.12186
The European Journal of Combinatorics, Volume 127, 104158

[7] *Constructing maximal pipedreams of double Grothendieck polynomials*
with Tianyi Yu
arxiv.org/abs/2312.01250
The Electronic Journal of Combinatorics, Volume 31, Issue 3

Conference Proceedings

[8] *Grothendieck polynomials of fireworks and inverse fireworks permutations*
with Linus Setiabrata and Tianyi Yu
Proceedings of the 39th Conference on Formal Power Series and Algebraic Combinatorics, extended abstract of [4] and [6].

Talks

Castelnuovo-Mumford regularity of skew-symmetric matrix Schubert varieties

- ALGECOM. University of Illinois, Urbana-Champaign. Nov. 2026
- Student Combinatorics & Algebra Seminar. University of Minnesota, Twin Cities. Sep. 2026

A positive combinatorial formula for the double Edelman–Greene coefficients

- Crystals, Schubert calculus, and Beyond. Joint Mathematics Meetings. Jan. 2027
- Algebraic Combinatorics and Applications. University of Notre Dame. Jul. 2026
- SIAM Math Slam. University of Florida. Apr. 2026
- 2026 Spring Southeastern Sectional of the AMS. Georgia Southern University. Mar. 2026
- Richard P. Stanley Seminar in Combinatorics. Massachusetts Institute of Technology. Mar. 2026
- Combinatorics Seminar. University of Minnesota, Twin Cities. Feb. 2026
- Student Combinatorics & Algebra Seminar. University of Minnesota, Twin Cities. Feb. 2026

Coxeter and Schubert combinatorics of μ–involutions

- Combinatorics Seminar. University of Kansas. Oct. 2026
- Illinois Summer Workshop in Algebraic Combinatorics. University of Illinois, Urbana-Champaign. Jun. 2026
- LACIM Seminar. LACIM, Université du Québec à Montréal. May 2026
- Discrete Geometry and Combinatorics Seminar. Cornell University. Feb. 2026
- Combinatorics Seminar. Center for Combinatorics, Nankai University. Jan. 2026
- Algebra Seminar. Beijing International Center for Mathematical Research, Peking University. Dec. 2025

Newton polytopes of fireworks Grothendieck polynomials

- Combinatorics and Graph Theory Seminar. Michigan State University (via Zoom). Sep. 2025
- Combinatorics Seminar. University of Florida. Sep. 2025
- New perspectives of coinvariant rings from viewpoints of geometry and algebraic combinatorics. RIMS, Kyoto University. Jul. 2025

Grothendieck polynomials of inverse fireworks permutations

- Algebra-Geometry-Combinatorics Seminar. University of Illinois, Urbana-Champaign. May 2025
- Combinatorics Seminar. University of California, San Diego. Mar. 2025
- Graduate Student Combinatorics Conference. University of Southern California. Mar. 2025

Constructing maximal pipedreams of double Grothendieck polynomials

- Student Combinatorics & Algebra Seminar. University of Minnesota, Twin Cities. Oct. 2026
- Combinatorics Seminar. National Taiwan Normal University. Dec. 2025
- Combinatorics Seminar. University of Florida. Oct. 2024
- Undergraduate Honor Thesis Presentation. University of California, San Diego. May 2024

Posters

Grothendieck polynomials of fireworks and inverse fireworks permutations

- Formal Power Series and Algebraic Combinatorics. University of Washington. Jul. 2026

Awards

University of Florida

- Joseph A. Alfred Mathematics Excellence Award (2000 USD) Sep. 2025

Others

- UCSD Undergraduate Honor Thesis (Honors with Highest Distinction):
  Three problems on Grothendieck polynomials and pipedreams

TEACHINGS

School of Mathematics, University of Minnesota, Twin Cities

- MATH 1271 Calculus I
  Fall 26

Department of Mathematics, University of Florida

- MAC 2312 Analytic Geometry and Calculus II
  Spring 26
- MAC 2311 Analytic Geometry and Calculus I
  Fall 24, Spring 25

PROFESSIONAL SERVICES

Referee for: Discrete Mathematics